

DengueWatch: A Digital Dengue Prevention and Community-Based Monitoring System

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Course: BS - Nursing 4D

Date: August 20, 2026

I. Introduction

Dengue remains a significant community health problem, particularly in areas where mosquitoes can easily reproduce in stagnant water found around homes and public spaces. The presence of uncovered containers, discarded materials, clogged drainage, and accumulated rainwater can provide potential breeding sites for mosquitoes. Communities may have difficulty identifying, reporting, and monitoring these areas consistently. With the increasing use of smartphones and digital platforms, technology can provide a practical way to improve community participation in dengue prevention. In this proposal, the author seeks to address the problem of dengue prevention by developing a digital monitoring system that allows community members to report suspected mosquito-breeding sites and enables health workers to monitor and respond to reported areas.

Problem Description

Dengue prevention requires continuous monitoring of possible mosquito-breeding sites within communities. However, some breeding sites may remain unnoticed or unreported, particularly when residents are unaware of their importance or when there is no organized system for reporting them. Stagnant water in containers, discarded tires, clogged drainage areas, and other water-holding materials can become potential breeding locations. During periods of frequent rainfall, these areas may increase and become more difficult to monitor manually.

Another challenge is the lack of a centralized system for recording and organizing reports of suspected breeding sites. Community members may notice stagnant water but may not know whom to contact or how to provide accurate information about its location. Health workers and local authorities may also find it difficult to determine which areas require immediate inspection when reports are scattered or recorded manually. This can delay community-based prevention activities and make monitoring less organized.

With this problem in mind, an accessible digital reporting and monitoring system can help improve the identification and management of potential mosquito-breeding sites. By allowing residents to submit reports with photographs and location information, community health workers can have a more organized source of information for inspection and follow-up.

Proposal Solution

The proposed solution to improve dengue prevention is the development of “DengueWatch”, a digital dengue prevention and community-based monitoring system. The system is designed to allow residents to report suspected mosquito-breeding sites through a mobile or web-based platform. Users can upload a photograph, provide a description, and submit the location of the suspected breeding site. This information can then be organized within the system for monitoring and follow-up.

One of the key features of DengueWatch is a *digital reporting system* that allows community members to submit reports conveniently. A *Dengue Hotspot Map* can also display reported locations, helping community health workers identify areas that may require inspection. Additional features include report status tracking, prevention reminders, educational information about dengue, and monitoring summaries showing the number of reported, inspected, and resolved sites.

The system primarily targets community residents, barangay health workers, local health authorities, and other individuals involved in dengue prevention. By connecting residents with community health workers through a centralized digital platform, DengueWatch encourages active community participation and improves the organization of dengue-related reports. The expected impact of this solution includes faster reporting of potential breeding sites, improved monitoring, increased awareness, and more organized dengue-prevention activities.

Overall, integrating digital reporting, location-based monitoring, and community participation provides a practical technology-enabled approach to supporting dengue prevention. The system can help communities organize information about potential breeding sites and support health workers in identifying areas that need attention.

Conclusion

Dengue prevention continues to require active participation from both individuals and communities because potential mosquito-breeding sites can develop in ordinary surroundings, especially when stagnant water is present. Lack of organized reporting and monitoring can make it difficult to identify and address these areas efficiently. Addressing this problem requires not only public awareness but also practical tools that encourage community involvement.

This proposal highlights the importance of implementing *DengueWatch*, a digital dengue prevention and community-based monitoring system. By allowing

residents to report suspected breeding sites, providing location-based monitoring, and organizing reports for health workers, the proposed system can support more systematic dengue-prevention activities. Ultimately, using technology to connect community reporting with local monitoring can promote greater awareness, faster response, and stronger community participation in dengue prevention.

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